

Grade 6 Accelerated
Unit 11
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

1. Joshua says that when he evaluated an expression, the result was 100. Use mental math to estimate each value. Then, circle which of the following expressions could have been Joshua's expression.

$$(10 + 2)^2 - 44$$

$$10^2 + 2^2 - 44$$

$$(10 - (-2)^2) + 44$$

$$12^2 + 44$$

2. Cindy evaluated the expression $3^4 \div 3(2.5 + 5.5)$.

Circle Cindy's error and choose which step of the order of operations she did not apply correctly.

Cindy's Steps	Order of Operations
$3^4 \div 3(2.5 + 5.5)$ $3^4 \div 3(8)$ $81 \div 3(8)$ $81 \div 24$ 3.375	• Evaluate expressions with exponents • Perform operations inside grouping symbols. • Perform multiplication and division in order from left to right. • Perform addition and subtraction in order from left to right.

3. Evaluate the expression without using a calculator. Write an explanation using the order of operations next to each step.

$ 7 - 10 \div (-6)$	Explanation
$ -3 \div (-6)$	grouping symbols
$-3 \div (-6)$	grouping symbols
$\frac{-3}{-6} \div 3 = -\frac{1}{2}$	Divide from left to right

4. Evaluate the expression without using a calculator. Write an explanation using the order of operations next to each step.

$(4 \times 1.5)^2 \div 9$	Explanation
$(6)^2 \div 9$	grouping symbols
$36 \div 9$	Evaluate expressions with exponents
4	Divide from left to right

Evaluate each expression.

5. $\frac{(-12 \times 0.25)^3}{9} = \frac{(-3)^3}{9} = \frac{-27}{9} = -3$

6. $5 \times |-8.8 \times 2.2| = 5 \times |-19.36| = 5 \times 19.36 = 96.8$

7. $(-92 \times 0.3)^2 + 52.24 = (-27.6)^2 + 52.24 = 761.76 + 52.24 = 814$

Determine if the equations in the table are true or false. Show work to support your answer.

Equation	True or False?
8. $3 - 3 \times 3^3 = -27$	$3 - 3 \times 27$ $3 - 81$ -78 True
9. $ 9.2 - 6.7 \times (-2) = -5$	$ -2.5 \times (-2)$ $2.5 \times (-2)$ -5 True
10. $5.1 \times 8.5 + (-4)^2 = 39.35$	$5.1 \times 8.5 + 16$ $43.35 + 16$ 59.35 False